

MIT ACADEMY OF ENGINEERING

COURSE CODE: CS104

5 APRIL 2024

FY BTECH SEMESTER-II 2022 (BACKLOG) PATTERN EXAMINATION 2023-2024

DEPARTMENT OF F.Y.BTECH
MID SEMESTER EXAMINATION
Essentials of Data Science

**SET
D**

TIME : 1 HOUR

MAX MARKS : 20

TOTAL NO OF QUESTIONS: 20

TOTAL NO OF PRINTED PAGES: 5

INSTRUCTIONS TO CANDIDATES:

1. Assume suitable data wherever necessary
2. Non programmable scientific calculators are allowed
3. Black figures to the right indicate full marks

- | | | | | |
|---|---|-----|-----|----|
| 1 | What does the expression $10 == 10$ and $5 > 6$ evaluate to? | [1] | CO1 | L2 |
| | a) TRUE b) FALSE c) Error d) None | | | |
| 2 | What will be printed by the following code when it executes?
<pre>i = 6
while i > -1:
 i = i - 1
print(i)</pre> | [1] | CO1 | L3 |
| 3 | How many asterisks will be printed when the following code executes?
<pre>for x in [0, 1, 2, 3]:
 for y in [0, 1, 2, 3, 4]:
 print('*')</pre> | [1] | CO1 | L3 |
| | a) 4 b) 5 c) 20 d) Error | | | |
| 4 | What value is printed when the following code is executed?
<pre>name = "Jane Doe"
def myFunction(parameter):
 value = "First"
 value = parameter
 print (value)

myFunction("Second")</pre> | [1] | CO1 | L3 |

	a) parameter	b) Second	c) value	d) First			
5	Which of the following is a disadvantage of recursive function?				[1]	CO1	L2
	a) It consumes more memory						
	b) Looks complex						
	c) Not easy to generate sequences						
	d) All the above						
6	Which of the following operator has its associativity from right to left?				[1]	CO1	L1
	a) +	b) //	c) %	d) **			
7	What will be printed by the following code when it executes?				[1]	CO1	L3
	<pre>i=0 while i<1: j=0 while j<4: print('*',end="") j = j+1 print('*') i = i+1</pre>						
8	What is the correct way to declare a global variable in Python?				[1]	CO1	L1
	a) global variableName			d) declare variableName			
	b) variableName = value at the top of the script						
	c) global variableName = value						
9	What will be printed by the following code when it executes?				[1]	CO1	L2
	<pre>def f1(x): global x x+=1 print(x) f1(15) print("hello")</pre>						
	a) 16	b) 16 hello	c) hello	d) error			

- 10 What will be printed by the following code when it executes? [1] CO1 L3
- ```
f = None
for i in range (5):
 with open("data.txt", "w") as f:
 if i > 2:
 break
print(f.closed)
```
- a) FALSE                      b) Error                      c) TRUE                      d) None
- 11 Which of the following is the correct way to create an empty tuple? [1] CO2 L1
- a) empty\_tuple = ()                      b) empty\_tuple = []  
c) empty\_tuple = {}                      d) empty\_tuple = ([])
- 12 Which of the following statements about sets in Python is true? [1] CO2 L1
- a) Sets maintain the insertion order of elements  
b) Sets are immutable  
c) Sets are indexed.  
d) Sets are mutable.
- 13 Which function checks if a specified element is present in a 2D list? [1] CO2 L2
- a) is\_present()                      b) in()  
c) exists()                      d) contains
- 14 What will be printed by the following code when it executes? [1] CO2 L3
- ```
dictionary = {'apple': 2, 'banana': 4, 'cherry': 6}
result = 0
for key in dictionary:
    result += dictionary[key]
print(result)
```
- 15 What will be printed by the following code when it executes? [1] CO2 L3
- ```
my_dict = {'a': 1, 'b': 2, 'c': 3}
result = 'a' in my_dict
print(result)
```
- a) FALSE                      b) TRUE                      c) Error                      d) 1

- 16      What will be printed by the following code when it executes?      [1]      CO2      L3
- ```
rows=[ ]
columns=[ ]
for i in range(4):
    for j in range(4):
        columns.append(j)
    rows.append(columns)
print(rows)
```
- 17 College name of student can be considered as an attribute of type? [1] CO2 L1
- a) nominal b) ordinal
c) interval d) ratio
- 18 What will be printed by the following code when it executes? [1] CO2 L3
- ```
people = {1: {'Name': 'John', 'Age': '27', 'gender': 'Male'},
 2: {'Name': 'Marie', 'Age': '22', 'gender': 'Female'}}

for p_id, p_info in people.items():
 print("\nPerson ID:", p_id)

 for key in p_info:
 print(key + ': ', p_info[key])
```
- 19      Which is the example of discrete attribute?      [1]      CO2      L2
- a) thickness of book in cm      b) weight of book in kg  
c) number of words in a book      d) book name
- 20      Which of the following is not a application for data science?      [1]      CO2      L2
- a) Mobile Recommendation System  
b) Temperature Prediction  
c) Stock Market Forecasting  
d) Cryptography